

Numerical Methods in Soft Matter Exercises

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Interaction potentials and thermostats

10.1 Canonical fluctuations

The exercise asked to show that the temperature fluctuations in the canonical ensemble are

$$\frac{\sigma_{T_K}^2}{\langle T_K \rangle^2} = \frac{\langle T_K^2 \rangle - \langle T_K \rangle^2}{\langle T_K \rangle^2} = \frac{2}{3N}, \quad (10.1.1)$$

where the kinetic temperature T_K is given by

$$T_K = \frac{1}{3Nk_B} \sum_{i=1}^N m_i v_i^2. \quad (10.1.2)$$

First, denoting the kinetic energy with E_K , we can rewrite T_K as

$$T_K = \frac{2E_K}{3Nk_B}, \quad \text{with } E_K = \frac{1}{2} \sum_{i=1}^N m_i v_i^2. \quad (10.1.3)$$

At this point we need the variance of E_K : we can start from the equipartition theorem

$$\langle E_K \rangle = \frac{3N}{2\beta}, \quad (10.1.4)$$

and being in the canonical ensemble we can derive with respect to β :

$$\sigma_{E_K}^2 = \langle E_K^2 \rangle - \langle E_K \rangle^2 = -\frac{\partial}{\partial \beta} \langle E_K \rangle = -\frac{3N}{2} \frac{\partial}{\partial \beta} \frac{1}{\beta} = \frac{3N}{2\beta^2}. \quad (10.1.5)$$

Putting this into eq. (10.1.3) we get

$$\sigma_{T_K}^2 = \langle T_K^2 \rangle - \langle T_K \rangle^2 = \frac{4}{9N^2 k_B^2} \frac{3N}{2\beta^2} = \frac{2}{3N k_B^2 \beta^2}. \quad (10.1.6)$$

By squaring eq. (10.1.3) we also get $\langle T_K \rangle^2$, so, finally:

$$\frac{\sigma_{T_K}^2}{\langle T_K \rangle^2} = \frac{2}{3N k_B^2 \beta^2} \frac{9N^2 k_B^2}{4 \langle E_K \rangle^2} = \frac{3N}{2\beta^2} \frac{4\beta^2}{9N^2} = \frac{2}{3N}. \quad (10.1.7)$$